EE5203 Microprocessors — Syllabus

2026-2027 · Istanbul Kültür University · Electrical & Electronics Engineering

What this course is

EE5203 develops embedded-system competence through a C-first STM32 pathway. You first build reliable Embedded C foundations, then learn enough register-level programming to understand the machine, and finally use the STM32 HAL to implement increasingly complex peripherals and an integrated project of your own.

Four ideas shape every week:

1. **Embedded C comes first.** Four full lectures build the C you need; everything later assumes it.
2. **Register literacy, HAL fluency.** You implement GPIO directly on registers once, so that when you use HAL afterwards you know what it does.
3. **Labs fade their scaffolding.** Early labs walk you through the workflow; later labs hand you a specification and expect a working, explained result.
4. **The project starts early.** You see the destination in Lecture 1, form teams mid-term, and demonstrate working subsystems as milestones — not one big demo at the end.

Learning outcomes

By the end of the course, a successful student can:

1. Write, compile, debug, and test Embedded C programs.
2. Explain the Cortex-M memory, register, and exception model.
3. Configure GPIO, timers, PWM, ADC, UART, and a synchronous serial peripheral using an appropriate combination of CMSIS register access and STM32 HAL.
4. Design interrupt-driven, non-blocking embedded applications.
5. Interpret datasheets, reference manuals, timing diagrams, compiler messages, debugger state, and peripheral registers as engineering evidence.
6. Integrate, test, document, demonstrate, and individually explain an STM32-based embedded system.

Tools and hardware

Item	What we use
Board	ST Nucleo-F401RE (provided in the laboratory)
Configuration	STM32CubeMX (pin configuration and project generation)
Development	VS Code with the course STM32 profile (edit, build, flash, debug)
Where things run	Laboratory PCs are fully set up; personal installation instructions are on the L01 lab reference page

Weekly rhythm

Each lecture package on this site has the same parts, used in the same order:

1. **Prepare** — a short study task and the pre-lab quiz (taken in CATS, two attempts, highest score counts).
2. **Learn** — the theory session and its study sheet.
3. **Practise** — a 30-45 minute worksheet; worked solutions are released in CATS after the corresponding session.
4. **Lab** — the lab guide, with an individual checkoff and a short oral check. Every lab follows the same cycle: warm-up, foundation checkpoint, independent core task, individual modification, oral check.
5. **Review** — a mastery checklist and exam-style questions inside the study sheet.

Graded pre-lab quizzes and homework are **only** in CATS — they are not published on this site. Announcements, submissions, and the weekly calendar schedule also live in CATS.

Course sequence

Thirteen lectures across a fourteen-week term. Week 8 is the midterm and introduces no new material.

ID	Week	Topic	Central performance	Project step
L01	1	Embedded systems, C toolchain, and debugging	Build, flash, debug, and modify Blinky	See example projects

ID	Week	Topic	Central performance	Project step
L02	2	Values, decisions, loops, and arrays	Process a sensor-data array in one C file	Identify project inputs
L03	3	Functions, arrays, strings, and modular C	Decompose the sensor program into functions and files	Identify processing functions
L04	4	Pointers, structures, bits, registers, and GPIO	Control PA5 through CMSIS registers	Individual idea canvas
L05	5	GPIO and hardware abstraction	Rebuild GPIO with HAL and compare the two interfaces	Teams form; project launch
L06	6	Interrupts and the NVIC	Implement and explain an event-driven application	Proposal and block diagram
L07	7	Timers and non-blocking time	Generate periodic events without blocking delays	Design review
	8	**Midterm examination**		
L08	9	PWM, output compare, and input capture	Generate timed signals — and measure them	First hardware milestone
L09	10	ADC and sensor acquisition	Acquire, scale, and validate analog data	Data-acquisition milestone

ID	Week	Topic	Central performance	Project step
L10	11	UART and PC communication	Stream acquired data to a PC and accept commands	Communication milestone
L11	12	I2C and SPI	Interface a synchronous serial device	Sensor/display integration
L12	13	System integration and project clinic	Integrate peripherals and diagnose faults	Working prototype
L13	14	Demonstration and defense	Demonstrate, test, and individually explain the system	Final assessment

Week numbers are the planned sequence; the authoritative weekly calendar, including any holiday shifts, is always the one announced in CATS.

Assessment

Assessment combines a midterm (including an individual toolchain practical), laboratory performance (individual checkoffs plus pre-lab quizzes and homework), the term project (proposal, milestones, integration, demo, report, and oral defense), and a final examination. Official weights follow the ECTS course record; any clarifications are announced in CATS.

Laboratory credit reflects **directly observed performance**: checkoffs are individual, and every lab ends with a short oral check where you explain your own work.

Project timeline

Point	Week	Milestone
L04	4	One-page individual idea canvas
L05	5	Teams form; preliminary topic

Point	Week	Milestone
L06	6	Proposal: problem, block diagram, components, minimum viable system
L07	7	Design review: pins, peripherals, data flow, responsibilities, risks
L08	9	First hardware milestone — one input or output peripheral working
L09	10	Acquired and validated sensor data
L10	11	Communication milestone — UART logging or command interface
L11	12	Integration milestone — input, processing, output, communication connected
L12	13	Readiness review: working prototype and test evidence
L13	14	Demonstration, report, presentation, and individual oral defense

There is no milestone in week 8. That week is yours for the midterm.